

An Experimental Analysis of Windows Log Events Triggered by Malware

Arpan Man Sainju
University of Alabama
P.O. Box 870290
Tuscaloosa, Alabama 35487
asainju@crimson.ua.edu

Travis Atkison*
University of Alabama
P.O. Box 870290
Tuscaloosa, Alabama 35487
atkison@cs.ua.edu

ABSTRACT

According to the 2016 Internet Security Threat Report by Symantec, there are around 431 million variants of malware known. This effort focuses on malware used for spying on user's activities, remotely controlling devices, and identity and credential theft within a Windows based operating system. As Windows operating systems create and maintain a log of all events that are encountered, various malware are tested on virtual machines to determine what events they trigger in the Windows logs. The observations are compiled into Operating System specific lookup tables that can then be used to find the tested malware on other computers with the same Operating System.

KEYWORDS

Digital Forensics, Windows Log Events, Survey

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1 INTRODUCTION

Malware detection is an important aspect of digital forensics. Blaming malware for evidence found on a defendant's PC is one of many tactics used by defense attorneys. In such cases, the judge cannot approve the infected PC as a reliable source of evidence. For example, in the State of Connecticut v. Julie Amero, Julie's conviction of impairing the morals of a child because her school computer started showing pornographic images was overturned on appeal when it was discovered that NewDotNet, malware capable of spawning uncontrollable pop-ups, had been installed on her computer 5 days prior to the alleged crime [1]. Thus, it is critical to be able to determine whether a PC contains malware capable of performing criminal or unauthorized activity.

*Corresponding author.

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The Event Logs in the Windows operating system, which record the activities performed in the system based on the enforced auditing policies, are considered a good source of digital information when looking for potential attacks on a system [19]. Classification of events is done based on the type. For example, information events can be the successful installation of software, warning events can be low disk space or possible malware detection, error events can be failed to install or load a piece of software [9]. Listing of all Event IDs with details such as type of event, the process of creation, Event name, etc., that are related to virus activities would certainly help digital forensics investigators.

1.1 Windows Logs

Windows Logs store the event record from legacy applications such as DOS applications, and events that apply to the system as a whole [30] in a file with .EVTX extension (.EVT for XP) [16]. These logs consist of event logs related to Application, Security, and System Logs. The Application Log records the events specific to computer programs or applications [30], i.e. event log for database update. The Security Log records the events related to login attempts, hardware resource utilization, etc. Finally, the System Log consists of events logged by Windows system components which are predefined by windows, such as the failure of a driver to load during startup [30].

Newer versions of Windows, Windows Vista and beyond, also include ForwardEvents and Setup Logs. All the events that are collected from remote computers are contained in the ForwardEvents Log. Creation of an event subscription is necessary for the system to collect an event record from remote computers [30]. The event subscription specifies the time in which the event from remote computers need to be collected and which log will be used to store the event locally [31]. The Setup Log consists of all the events related to an application setup [30], i.e. database software setup.

1.2 Application and Service Logs

Starting with Windows Vista, Windows provided a new category of logs: Application and Service Logs. These logs store events from a single application or component rather than events that might have a system-wide impact [30], and consist of admin, operational, analytic and debug logs. The events in the Admin Log are for end users such as administrators, and provide a well-defined solution to go along with the problem indicated by the event [30]. For instance, an event that shows a record of failing to connect to a printer will also have a message with instructions to fix the problem [30].

Operational events are used for analyzing and diagnosing a problem or occurrence [30]. These events can be used to start a tool or a task related to the problem or occurrence [30]. The Analytic Log,

however, is hidden from the user. It records the program operation events and indicates problems that cannot be handled by the user [30]. Finally, the Debug Log is used by the developer to troubleshoot problems with their programs.

By using the logs provided by the Windows operating system, this effort will try to identify the events that occur after malware is installed. Section 2 will introduce some of the malware which are of interest to digital forensics. Related works will be discussed in Section 3. Section 4 describes the experiment setup. Section 5

presents results based on the experiment and the conclusion is presented in Section 6.

2 MALWARE

Malware is software designed to gain access to a computer without the knowledge of the owner [5]. There are many types of malware including viruses, worms, spyware, ransomware, keyloggers, and trojan horses [5]. A computer virus spreads by attaching to legitimate executable files and infecting hosts through email attachments, downloads, or removable media [28]. In contrast, computer worms

Table 1: Malware

Malware	Type	Description
Graybird	Trojan	Provides unauthorized access to the infected computer which allows the Trojan creator to get information about the system and network, install an FTP server to use the infected computer as a temporary storage device, or download and execute files. Also accesses password cache and interprets keystrokes to steal information. [22]
Agobot	Trojan	Also known as Gaobot [10]. Remotely bypasses normal security mechanisms of the infected computer. May also contain features such as a packet sniffer, keylogger, rootkit installer, or DDoS attack. [2]
Zeus	Trojan	Uses phishing attacks to infect the computer. Performs malicious activities such as stealing banking information or installing ransomware [4] [18].
Kelz	Worm	Propagates through emails. Copies itself to local, mapped, and network drives, then looks for email addresses stored in various locations of the infected system. Using its own SMTP engine, it sends an email, with the worm attached, from a random email address found on the infected computer to the other addresses found. The subject line, message body and attachment filename are also random[11]. Since Kelz randomly chooses the "From" address, many cases were reported blaming uninfected users for sending malicious emails [11].
ZeroAccess	Trojan	Uses an advanced rootkit to hide and can create a hidden file system, download more malware and open a backdoor on the infected computer [23]. It is mostly involved in downloading other malware on an infected system for bitcoin mining or click fraud [3]. Uses social engineering techniques such as phishing attacks for infecting the system [3].
Salaty	Worm	Infects executable files, creates a P2P botnet and receives URLs for downloading other files in the infected PC [32].
Mdropper	Trojan	Exploits Microsoft Word or Excel vulnerabilities to drop other malware on the infected computer [24]. Usually used by the attacker to hide their malware, since it presents itself as an authentic Word or Excel file.
Bladabindi	Trojan	Steals confidential information from the infected computers [13].
Vobfus	Worm	Also known as Changeup worm. Exploits the Microsoft Windows Shortcut 'LNK' File's Automatic File Execution vulnerability. Shows misleading applications as well as downloads other threats on the infected PC.
Refog	Keylogger	Logs all keystrokes and monitors all applications and resources used.
Kidlogger	Keylogger	Logs all keystrokes and monitors all applications and resources used.

Table 2: Result of Experiment in Window 7

Malware	Event Source	Event ID	Event Description	Type	Log
Klez	DNS Client Event	1014	Name resolution for the name smtp.ONLINE.SH.CN timed out after none of the configured DNS servers responded.	Warning	System
ZeroAccess	Service Control Manager	7040	The start type of the Windows Defender service was changed from auto start to disabled.	Information	System
	Service Control Manager	7040	The start type of the IP Helper service was changed from auto start to disabled.	Information	System
	Service Control Manager	7040	The start type of the Security Center service was changed from auto start to disabled.	Information	System
	Service Control Manager	7040	The start type of the Security Center service was changed from auto start to disabled.	Information	System
	Service Control Manager	7040	The start type of the Base Filtering Engine service was changed from auto start to disabled.	Information	System
	Service Control Manager	7036	The Windows Firewall service entered the stopped state.	Information	System
	Service Control Manager	7036	The Security Center service entered the stopped state.	Information	System
	Service Control Manager	7036	The IP Helper service entered the stopped state.	Information	System
	Service Control Manager	7036	The Base Filtering Engine service entered the stopped state.	Information	System
	Service Control Manager	7036	The Windows Defender service entered the stopped state.	Information	System
	SecurityCenter	2	The Windows Security Center Service has stopped.	Information	Application
Salaty	Application-Experience	201	The Program Compatibility Assistant service started successfully.	Information	System
	Application-Experience	206	The Program Compatibility Assistant service successfully performed phase two initialization	Information	System
MDropper	Application-Experience	201	The Program Compatibility Assistant service started successfully.	Information	System
	Microsoft Windows Security Auditing	6424	An account was successfully logged.	Information	Security
	Microsoft Windows Security Auditing	4672	Special privileges assigned to new logon.	Information	Security
Bladabindi	Service Control Manager	7040	The start type of the Background Intelligent Transfer Service was changed from demand start to auto start.	Information	System
Vobfus	Schannel	36887	The following fatal alert was received: 40.	Information	System
Refog Keylogger	Service Control Manager	7045	A service was installed in the system. Service Name: Local Synchronization Host ..	Information	System
	Service Control Manager	7036	The Local Synchronization Host service entered the running state.	Information	System
kidlogger	Service Control Manager	7045	A service was installed in the system. Service Name: Monitoring agent	Information	System
	Service Control Manager	7036	The Local Synchronization Host service entered the running state.	Information	System

can replicate and spread to other computers without attaching to an existing program [6] [12]. Spyware is used to steal confidential and credential information [5]. Ransomware is designed to block access to a computer system until a sum of money is paid [7]. A keylogger, such as Refog Keylogger or Kidlogger, logs all the keystrokes pressed by the user, and is used to steal information such as passwords. [20]. A Trojan Horse is designed to gain unauthorized remote access to the infected system by hiding itself in a legitimate program [29]. Also, software backdoors can be used to gain unauthorized access to a computer by using an installed program or rootkit which is capable of bypassing normal authentication [27].

This effort focuses on malware used for confidential information stealing, unauthorized remote access, and bypassing authentication mechanisms [26] such as Trojan Horses, keyloggers, and spyware. The malware described in Table 1 were selected based on their sample availability through sources such as *vxheaven.org* and *offensivecomputing.net* as well as their relevance to digital forensics. They were also chosen because each one targets Windows-based operating systems.

3 RELATED WORKS

Various work has been done in analyzing the effect of malware on a system based on various requirements. Distiller et al. [8] performed malware analysis for creating an incident response system that addresses what to do after an attack is performed on the system. Their main goal was to handle malware attacks in a way that restricts the damage and minimizes the time and cost to recover. They utilize a virtual machine for analyzing the malware because of its feature that allows installation of various Operating Systems as an application in a single host machine. To analyze the behavior of the malware, they used tools such as BgInfo, Process Explorer, Proces Monitor, PSfile, and RootkitRevealer to analyze registry files, monitor processes, search ANSI and UNICODE string in binary images, port scanning, and scanning disks to retrieve hidden files. However, they did not include analyzing Windows Event Log files.

Windows Log Events are being used to extract the machine reboot event and classify the cause of the reboot [25]. The cause classification and availability analysis given by the reboot information provides valuable insight in dependability of the system. Similarly, Sahoo et al. [21] performs the real-time evaluation of event logs for diagnosing as well as prediction of node conditions in distributed computing environments.

Windows Log Events are also being used for malware analysis. Mullinix [17] includes analysis of Microsoft Event Logs to determine whether a virus has been installed on a target system or not. The author includes analysis of three viruses (Fizzer, Zeus and MyDoom), and creates a table that records event IDs that suggest a malware installation on a system. However, they do not consider the specific event description and event source. Two different event sources may have the same event ID which may specify different events; therefore, the table could be ambiguous. Also, since the event with the same event ID and event source can have different event descriptions and the table records only the generic description of an event, exact information about the event is lost. For example, generic event description may include "Window service fails", but

specific event description specifies which service fails such as "Security Center service was stopped". These issues will be addressed in our experimental result.

4 EXPERIMENT SETUP

A virtual machine allows the creation of a virtual environment based on the operating system that is installed on it, and has been extensively used for quick, cost-effective testing of newly developed tools in different operating system environments. Likewise, virtual machines can also be used to safely study the effects of malware in different operating system environments without having to completely wipe the machine after experimentation. The virtual machine can simply be removed from the host machine. Hence, the experiments in this paper will be using a virtual machine to study the malware. In this paper, VMware Workstation 12 player was used to create virtual machines with a fresh copy of Windows XP/Windows 7 installed on them.

Each Virtual Machine was configured with two CPU processors, 512 MB of memory and 40GB of hard drive storage. Separate virtual machines were built to test sample malware individually. After the completion of each test, respective virtual machines were immediately deleted from the host machine.

The malware under investigation were downloaded from the following sources: *offensivecomputing.net*, *malwr.com*, and *vxheaven.org*. Those sources provide the sample malware based on their digital signature, hash value, and type. The malware samples are zipped and password protected.

The experiment starts with the downloaded zipped file of the malware inside the virtual machine. The malware is extracted from the downloaded file and run on the system to infect. Any warning pop-ups that appear are ignored to ensure the malware infects the system. The next step is to note which event is triggered due to the malware that is currently affecting the system through the Event Viewer. Similarly, each malware was installed onto different virtual machines, according to their compatibility (Windows XP and/or Windows 7), to analyze what set of events are triggered by each of them.

5 RESULTS

The observations from the experiment discussed above were compiled in lookup tables. Table 2 is based on observations of malware activities in Windows 7; however, due to space restrains, the lookup table for Window XP is not included.

An event with the same source and event ID may specify different events. For example, the event of Refog Keylogger malware with event source "Service Control Manager" and event ID 7036 had the event description "The Local Synchronization Host service entered the running state"; however, in the case of ZeroAccess, there are 5 different events with the event source "Service Control Manager" and event ID 7036 and each has a different event description. Therefore, it is crucial to note the event description of the triggered events. Moreover, certain event descriptions may be specific to a certain piece of malware. For example, when testing on Windows XP, the Kelz malware triggered an event with the description "The Krn132 service was successfully sent a start

control". The krnl32 service starting is specific to Kelz malware, i.e. it ensures that Kelz malware was installed on the machine [15].

The event triggered after a malware is installed on a system is Operating System specific, i.e. the events that are recorded after the installation of the same malware in Windows XP and Windows 7 are different. For example, ZeroAccess malware in Windows XP triggered events such as stopping Windows Firewall/Internet Connection Sharing (ICS) service, Security Center service, and Application Layer Gateway service. However, in Windows 7, it triggered events such as stopping Windows Firewall service, IP Helper service, Security Center service, Windows Defender and Base Filtering Engine service. Therefore, separate Operating System specific lookup tables were created.

Experiments were conducted with two different open source keyloggers: Refog keylogger and Kidlogger. Both of the keyloggers had a common event of starting the Local Synchronization Host service. It suggests that running of Local Synchronization Host service may imply installation of a keylogger in the machine.

Some of the events recorded in Windows logs after a malware installation can be common events that may also be triggered from non-malicious programs. For example, one of the Events recorded after installation of Agobot in Windows XP was the event with Event ID: 4097 and Event Source: Dr. Watson. This event usually occurs when an application, whether malware or not, tries to access unreachable or unavailable main memory locations. Nonetheless, such recorded events provide a strong hint that some kind of malicious program may be installed on the PC, and hence, indicate the need for a more complex digital forensic investigation.

Comparisons between the lookup tables from the experiment and raw log files, based on common attributes like Event ID, Event Source, and Event Description, can be utilized to quickly extract the records in log files which suggest the possibility of malware installation. If no records match after the comparison, it suggests that the machine is clean, i.e. no malware found. Hence, our lookup table can be used to get a quick and early notion of whether a malicious program was installed on the PC. Also, considering that the default maximum size of log files for most Windows operating systems is only around 4.2 GB, execution of the comparison between the log file and the respective lookup table may not take longer than a few minutes. A tool like Log Parser GUI [14], created by Microsoft, provides us the capability to perform SQL queries to compare the log files.

6 CONCLUSION

In conclusion, before making a judgment on a court case that involves digital evidence, it is essential to first ensure that the digital media is not compromised by any kind of malware. The lookup table from our experiment would be beneficial to extract evidence on malware installation. Furthermore, this lookup table is capable of determining which malware was installed on the suspect's Windows machine. Also, such a lookup table would certainly be helpful because an infected machine may not be considered as a source of evidence. It would help digital media forensics investigators to easily reject such devices and save their time and energy.

Finally, considering the above-mentioned benefits, this project requires the contribution from digital forensics investigators and

enthusiasts for the table to include the reference on triggered event logs after the installation of all the digital forensics relevant malware in its respective Windows based Operating System.

REFERENCES

- [1] 2004. State of Connecticut v. Amero case 2004. (2004). https://en.wikipedia.org/wiki/Connecticut_v._Amero.
- [2] 2016. Agobot. (2016). <https://en.wikipedia.org/wiki/Agobot>.
- [3] 2016. ZeroAccess botnet. (2016). https://en.wikipedia.org/wiki/ZeroAccess_botnet.
- [4] 2016. Zeus (malware). (2016). [https://en.wikipedia.org/wiki/Zeus_\(malware\)](https://en.wikipedia.org/wiki/Zeus_(malware)).
- [5] John Aycock. 2006. *Computer viruses and malware*. Vol. 22. Springer Science & Business Media.
- [6] M. Barwise. 2010. What is an internet worm? (2010). <http://www.bbc.co.uk/webwise/guides/internet-worms>.
- [7] Malware Protection Center. 2016. Ransomware. (2016). <https://www.microsoft.com/en-us/security/portal/mmpc/shared/ransomware.aspx>.
- [8] Dennis Distler and Charles Hornat. 2007. Malware analysis: An introduction. *SANS Institute InfoSec Reading Room* (2007).
- [9] Toshihisa Fujii, William L. Call, and Abhijit P. Rane. 1999. Error management system for supporting the identification and logging of error messages. (April 6 1999). US Patent 5,892,898.
- [10] S. Gettis. 2003. Backdoor.Gaobot. (2003). https://www.symantec.com/security_response/writeup.jsp?docid=2003-080607-5215-99.
- [11] A. Gudmundsson. 2002. W32.Kelz.E@mm. (2002). https://www.symantec.com/security_response/writeup.jsp?docid=2002-011716-2500-99.
- [12] Lorine A. Hughes and Gregory J. DeLone. 2007. Viruses, worms, and trojan horses: Serious crimes, nuisance, or both? *Social science computer review* 25, 1 (2007), 78–98.
- [13] S. Karpe. 2013. Trojan.Bladabindi. (2013). https://www.symantec.com/security_response/writeup.jsp?docid=2013-072415-3728-99.
- [14] Lizard Labs. 2016. Log Parser Lizard. (2016). http://www.lizard-labs.com/log_parser_lizard.aspx.
- [15] Malwarebytes. 2016. Pacman's Portal. (2016). <http://www.pacs-portal.co.uk>.
- [16] Robert A. Milford. 2011. Method for transforming and consolidating fields in log records from logs generated on different operating systems. (Dec. 27 2011). US Patent 8,086,650.
- [17] Michelle D. Mullinix. 2013. *An Analysis of Microsoft Event Logs*. Ph.D. Dissertation. Utica College.
- [18] B. Nahorney and N. Falliere. 2010. Trojan.Zbot. (2010). https://www.symantec.com/security_response/writeup.jsp?docid=2010-011016-3514-99.
- [19] L. Rocha. 2015. *Digital Forensics - Supertimeline and Event Logs*. <https://countuponsecurity.com/2015/11/23/digital-forensics-supertimeline-event-logs-part-i/>.
- [20] Seref Sagiroglu and Gurol Canbek. 2009. Keyloggers. *IEEE technology and society magazine* 28, 3 (2009), 10–17.
- [21] Ramendra K. Sahoo, Adam J. Oliner, Irina Rish, Manish Gupta, José E. Moreira, Sheng Ma, Ricardo Vilalta, and Anand Sivasubramaniam. 2003. Critical event prediction for proactive management in large-scale computer clusters. In *Proceedings of the ninth ACM SIGKDD international conference on Knowledge discovery and data mining*. ACM, 426–435.
- [22] S. Sevcenco. 2003. Backdoor.Graybird. (2003). https://www.symantec.com/security_response/writeup.jsp?docid=2003-040217-2506-99.
- [23] J. Shearer. 2011. Trojan.Zeroaccess. (2011). https://www.symantec.com/security_response/writeup.jsp?docid=2011-071314-0410-99.
- [24] J. Shearer and H. Lau. 2005. Trojan.Mdropper. (2005). https://www.symantec.com/security_response/writeup.jsp?docid=2005-031911-0600-99.
- [25] Cristina Simache, Mohamed Kaâniche, and Ayda Saidane. 2002. Event Log based Dependability Analysis of Windows NT and 2K Systems.. In *PRDC*, Vol. 2. Citeseer, 311.
- [26] B. Stillinger. 2013. The Importance of Digital Forensics. (2013). <http://investigatesc.com/importance-digital-forensics/>.
- [27] Symantec. 2016. Software Backdoor. (2016). <http://www.pctools.com/security-news/software-backdoors/>.
- [28] Symantec. 2016. What do computer virus do? (2016). <http://www.pctools.com/security-news/what-do-computer-viruses-do/>.
- [29] Symantec. 2016. What is Trojan Virus? (2016). <http://www.pctools.com/security-news/what-is-a-trojan-virus/>.
- [30] Technet.microsoft.com. 2016. Event Logs. (2016). [https://technet.microsoft.com/en-us/library/cc722404\(v=ws.11\).aspx](https://technet.microsoft.com/en-us/library/cc722404(v=ws.11).aspx).
- [31] Technet.microsoft.com. 2016. Event Subscriptions. (2016). <https://technet.microsoft.com/en-us/library/cc749183.aspx>.
- [32] A. Thigpen and E. Chien. 2006. W32.Sality. (2006). https://www.symantec.com/security_response/writeup.jsp?docid=2006-011714-3948-99.